Optimal obstacle regularity for stable operators with minimally smooth obstacles

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Abstract

We prove the interior C^{1+s} estimate for obstacle problems driven by symmetric stable operators with positive angular Lipschitz density, for every obstacle in C^β with noninteger $\beta > 1+s$. The exterior values of the solution are arbitrary bounded data. This gives the minimally smooth obstacle threshold in the local anisotropic problem. After subtracting and localizing the obstacle, the forcing need only be Hölder continuous. We compare the solution with shifted half-space profiles in a known positive half-ball. A thin-strip Dirichlet decomposition controls the constant forcing defect; a flat boundary quotient estimate gives the outer variation. Blow-ups at actual free-boundary points identify the quotient's signed trace, yielding improvement of the value bounds and of the compensated tail. A scale dichotomy and an interior covering argument give the full C^{1+s} norm estimate.

1 Introduction

The nonlocal obstacle problem asks for a function that lies above a prescribed obstacle and is harmonic for a nonlocal elliptic operator wherever it separates from that obstacle. It is also the stationary optimal stopping equation for a stable jump process, with the obstacle playing the role of its payoff; see [3, Chapter 4]. The set where the two functions meet is unknown. Regularity across its boundary is therefore stronger than an interior estimate in the harmonic region. The characteristic detachment profile has degree $1+s$ for an operator of order $2s$, making C^{1+s} the natural regularity to seek.

For the fractional Laplacian, Caffarelli, De Silva and Savin proved this regularity with obstacles in C^β for $\beta > 1+s$ [2, Theorem 6.1]. Their argument uses a truncated frequency formula for the extension equation. For general stable operators, the local result of Ros-Oton, Torres-Latorre and Weidner gives every subcritical estimate C^q, $q < 1+s$, and the endpoint C^{1+s} estimate when $\beta > 1+2s$ [4, Corollary 1.3]. Their estimates allow arbitrary bounded exterior values. The remaining minimally smooth anisotropic question is recorded in [3, Open Question 4.1, Section 4.6.1].

We prove the same local endpoint estimate for every $\beta > 1+s$ under angular Lipschitz control. More precisely, let

$$Lu(x) = \mathrm{PV} \int_{\mathbb{R}^n} (u(x) - u(x+y)) \frac{a(y/|y|)}{|y|^{n+2s}}\, dy. \tag{1.1}$$

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Theorem 1.1. *Let $n \geq 2$, $0 < s < 1$, and let $\beta > 1 + s$ be noninteger. Suppose $a \in \mathrm{Lip}(S^{n-1})$ is even and $0 < \lambda \leq a \leq \Lambda$. For every $\phi \in C^\beta(B_1)$ and every bounded continuous viscosity solution of*

$$\min\{Lu, u - \phi\} = 0 \quad \text{in } B_1,$$

one has

$$\|u\|_{C^{1+s}(B_{1/2})} \leq C \left(\|\phi\|_{C^\beta(B_1)} + \|u\|_{L^\infty(\mathbb{R}^n)} \right). \tag{1.2}$$

Here C depends only on $n, s, \beta, \lambda, \Lambda$ and $\|a\|_{\mathrm{Lip}(S^{n-1})}$.

Thus the obstacle, solution concept, operator sign and quantitative norm are those of the local question. The estimate includes arbitrary bounded exterior values and does not require the coincidence set to stay away from ∂B_1. Its new range relative to the cited anisotropic endpoint statement is $1 + s < \beta \leq 1 + 2s$.

The difficulty is visible after subtracting the obstacle. If its usable smoothness is $1 + s + \alpha$, the new forcing is only $C^{1-s+\alpha}$, below Lipschitz regularity when $\alpha < s$. The convexity and classification of blow-ups from [4, Theorem 1.5] still provide a coarse half-space profile. Turning this coarse information into an endpoint bound requires finer control of the contact layer. Merely locating the free boundary does not order the values above it.

We obtain that control from a value sandwich between two shifted profiles. It supplies a known positive half-ball above the thin uncertain layer. A Dirichlet decomposition on this fixed flat geometry separates the layer's harmonic influence from a part that vanishes on the flat boundary. Torsion at the layer scale bounds the unknown constant forcing defect. The quotient of the zero-exterior part by the distance to the flat boundary to the power s then has a first-order expansion. We prove the required flat quotient estimate using only angular Lipschitz regularity and the one-dimensional higher Liouville theorem of [1, Proposition 3.11].

The sign of the expansion is fixed by blowing up at actual free-boundary points inside the layer. Matching these inner limits to the outer quotient ties its trace to the true contact displacement. This also excludes positive pockets below the limiting contact plane. The resulting value comparison and tail improvement iterate. A first-failure choice of scale treats the nondegenerate branch, while the complementary branch already has stronger decay. Finally, derivatives of a localized solution satisfy linear equations in the positive phase, which converts the one-point estimate into (1.2).

The strict inequality $\beta > 1 + s$ supplies a positive Hölder margin in this argument. We do not address the equality case or nonhomogeneous kernels. Section 2 proves the flat boundary estimate; Section 3 proves improvement of flatness; and Section 4 completes the localization, scale selection and norm estimate. We use the positive operator convention (1.1); the norms and weighted tails are specified at their first use below.

2 Linear estimates in a flat half-space

Throughout the paper the operator has the positive sign convention

$$Lu(x) = \mathrm{PV} \int_{\mathbb{R}^n} (u(x) - u(x + y)) K(y)\, dy, \qquad K(y) = \frac{a(y/|y|)}{|y|^{n+2s}}, \tag{2.1}$$

where a is even, $\lambda \leq a \leq \Lambda$, and $\|a\|_{\mathrm{Lip}(S^{n-1})} \leq M$. Constants may depend on $n, s, \lambda, \Lambda, M$ and on the fixed auxiliary exponents. They do not depend on an individual solution's bound at infinity unless that bound is displayed. We write $x = (x', t)$ and $\mathbb{H} = \{t > 0\}$.

For noninteger $\nu = k + \theta$, $k \geq 0$ and $0 < \theta < 1$, C^ν has its usual inhomogeneous norm: the supremum norms of derivatives through order k, together with the θ-Hölder seminorm of the derivatives of order k. Brackets $[\cdot]_{C^\nu}$ denote the homogeneous seminorm. The weighted value norm used below is

$$\|w\|_{L^\infty_\mu} = \sup_{x \in \mathbb{R}^n} \frac{|w(x)|}{(1 + |x|)^\mu}.$$

All equations in this section can be read distributionally. At the regularity greater than $2s$ used in the nonlinear argument, this agrees with the classical integral and the corresponding viscosity equation.

2.1 Growing profiles and local estimates

The functions t_+^{1+s} and $x_i t_+^s$ are useful comparison functions, but their full integrals in (2.1) need not converge at infinity. We use their operator only up to an additive constant. More precisely, for a smooth radial cutoff χ_R, equal to one on B_R and zero outside B_{2R}, the assertion $LU = F$ modulo constants means that there are constants c_R such that

$$L(\chi_R U) - c_R \longrightarrow F$$

on compact subsets of the domain, in distributions. The estimates below also give convergence of the oscillatory far part. This is the $k = 1$ interpretation of [1, Definition 3.1].

Angular Lipschitz continuity gives, for $|y| > 2\max\{|x|, |z|\}$,

$$|K(x - y) - K(z - y)| \leq C|x - z|\,|y|^{-n-2s-1}. \tag{2.2}$$

Consequently, if $|U(y)| \leq A(1 + |y|)^{1+s+\tau}$ and $\tau < s$, then

$$\int_{|y|>R} |U(y)|\,|K(x - y) - K(z - y)|\,dy \leq CA|x - z|R^{\tau-s} \tag{2.3}$$

for x, z in a fixed ball and R sufficiently large. The radial integral is $\int_R^\infty r^{\tau-s-1}\,dr$. In particular, truncations can change the operator by a divergent constant, but their spatial oscillations converge.

We use the following local estimates from [3]; theorem numbering in that reference refers to its specified author manuscript. A weak solution with zero local exterior on a known flat boundary, bounded right-hand side and bounded weighted tail is locally C^s up to that boundary [3, Proposition 2.6.4]. For a continuous distributional solution the same assertion follows from [3, Proposition 2.6.13]. In the latter statement choose its boundary exponent to be $s/2$; a bounded right-hand side controls the required weighted right-hand side. These results are used on fixed flat charts, away from any corner of the auxiliary domain.

In the interior, a bounded cutoff solution of $Lw = F$ with $F \in C^\gamma$ is $C^{2s+\gamma}$, provided $2s + \gamma$ is noninteger and the kernel has the integrated C^γ regularity in [3, Definition 2.1.22] [3, Theorem 2.4.1]. Here every $0 < \gamma < 1$ is allowed by (2.2). The weighted-tail version will always be applied after the compensation just described.

There is one useful way of controlling a missing constant. Suppose $LU = 0$ modulo constants in a fixed half-ball and U has the growth in (2.3). For a fixed compact cutoff $U_0 = \chi U$, write

$$LU_0 = d + G,$$

where G has bounded Lipschitz oscillation, by (2.3). Normalize G to vanish at one interior point. Testing against a fixed smooth function ζ supported strictly inside the half-ball, with $\int \zeta = 1$, gives

$$d = \int_{\mathbb{R}^n} U_0 L\zeta - \int G\zeta. \tag{2.4}$$

Both terms are bounded by CA. Indeed $L\zeta \in L^1$, using the second difference of ζ near zero and integrability of K at infinity. Thus the flat C^s estimate applies with a controlled full right-hand side. This argument also applies to equations whose right-hand-side oscillation tends to zero.

These observations justify compactness with varying kernels. Uniformly Lipschitz angular densities have a uniformly convergent subsequence on the sphere. On finite annuli this convergence passes the operator to the limit; (2.3) controls the remaining tails. No convergence of angular derivatives is needed. Equivalently, one may test against compact smooth functions of integral zero: their images under L have the extra far decay $|y|^{-n-2s-1}$.

2.2 A higher half-space Liouville theorem

For one-variable functions, polar coordinates and evenness give

$$L(F(x\cdot e)) = A_e L_s F(x\cdot e), \qquad A_e = \frac{1}{2}\int_{S^{n-1}} a(\theta)|\theta\cdot e|^{2s}\, d\theta > 0, \tag{2.5}$$

where $L_s F(t) = \mathrm{PV}\int_{\mathbb{R}}(F(t) - F(t+y))|y|^{-1-2s}\, dy$. The same formula holds for compensated operator differences. In particular $L(t_+^s) = 0$ in $\mathbb{H}$ [3, Lemma 2.6.2 and Lemma B.1.5]. We will also use

$$L((a_0 + b\cdot x)t_+^s) = 0 \quad \text{modulo constants in } \mathbb{H}. \tag{2.6}$$

For the normal linear term, differentiate t_+^{1+s} spatially and use $L(t_+^s) = 0$. For a tangential index $i < n$, vary the unit normal along

$$e(\varepsilon) = \frac{e_n + \varepsilon e_i}{|e_n + \varepsilon e_i|}, \qquad \frac{d}{d\varepsilon}(x\cdot e(\varepsilon))_+^{1+s}\Big|_{\varepsilon=0} = (1+s)x_i t_+^s.$$

Use radial cutoffs of radius R first and differentiate their spatial operator oscillation with respect to this normal parameter. The derivative tail errors on compact sets are $O(R^{-s})$. In the directional calculation the factor $|\theta\cdot e|^{s-1}$ is integrable on the sphere. Differentiation of the finite integrals, followed by $R\to\infty$, shows that the spatial oscillation of the operator vanishes, which is exactly (2.6).

Lemma 2.1. *Let $0 < \tau < \min\{s, 1-s\}$. Suppose that U is continuous on $\mathbb{R}^n$, vanishes in $\{t \le 0\}$, solves $LU = 0$ modulo constants in $\mathbb{H}$, and satisfies $|U(x)| \le A(1+|x|)^{1+s+\tau}$. Then*

$$U(x) = (a_0 + b\cdot x)t_+^s.$$

Proof. Apply the cutoff estimate and (2.4) to $R^{-1-s-\tau}U(R\,\cdot)$, and then the flat boundary estimate. This gives

$$[U]_{C^s(B_R)} \le CAR^{1+\tau}, \qquad R \ge 1. \tag{2.7}$$

A tangential increment, divided by $|h|^s$, has growth at most $1+\tau$, vanishes on the lower half-space, and still solves the zero equation modulo constants. The same estimate can be applied repeatedly; the compensated-tail calculation remains valid as the growth decreases. For a fixed number j

with $js > 1+\tau$, the $C^s(B_R)$ seminorm of a j-fold tangential increment is bounded by $CR^{1+\tau-js}$. Sending R to infinity shows that this increment is constant, and its vanishing on the lower half-space makes it zero.

Thus $U(\cdot,t)$ is a polynomial in x'. For completeness, convolve in x', divide the vanishing finite differences by their step sizes, and let the steps tend to zero. The resulting smooth function has zero derivatives of order j; passing back through the convolution gives the assertion for the continuous function. Since $1+s+\tau<2$, its degree in x' is at most one. Write

$$U(x',t) = A_0(t) + \sum_{i=1}^{n-1} x_i B_i(t).$$

Evaluating this identity at $x' = (1+|t|)e_i$ and at $x' = 0$ bounds $|B_i(t)| \le CA(1+|t|)^{s+\tau}$. Tangential differences of the equation show that $LB_i = 0$ modulo constants. This operator is now an ordinary convergent integral, because $s+\tau<2s$, and its value is a constant. Scaling by R and testing inside the half-space bounds that constant by $CR^{\tau-s}$, so it is zero. The lower-growth half-space Liouville theorem [3, Theorem 2.7.2] gives $B_i(t) = b_i t_+^s$.

Subtract these tangential terms using (2.6). The remaining one-dimensional A_0 is zero for $t \le 0$, has growth $1+s+\tau$, and solves $L_s A_0 = 0$ modulo constants by (2.5). The one-dimensional higher Liouville theorem [1, Proposition 3.11, $k=1$] gives $A_0(t) = (a_0 + b_n t)t_+^s$. □

2.3 The quotient estimate

Proposition 2.2. *Fix $0<\alpha<\min\{s,1-s\}$, put $\gamma = 1-s+\alpha$, and let $0<\tau<\alpha$. If q is bounded on $\mathbb{R}^n$, continuous in B_1, and*

$$Lq = F \quad \text{in } B_1\cap\mathbb{H}, \qquad q = 0 \quad \text{in } B_1\cap\{t\le 0\}, \qquad F \in C^\gamma(\overline{B_1\cap\mathbb{H}}),$$

then q/t^s extends to $C^{1,\tau}(\overline{B_{1/2}\cap\mathbb{H}})$ and

$$\left\| \frac{q}{t^s} \right\|_{C^{1,\tau}(\overline{B_{1/2}\cap\mathbb{H}})} \le C(\|q\|_{L^\infty(\mathbb{R}^n)} + \|F\|_{C^\gamma}). \tag{2.8}$$

Only local vanishing on the lower half-space is assumed.

Proof. Normalize the right side to one. We first prove, at every flat boundary point z in a smaller ball, an affine approximation

$$\|q - Q_z t_+^s\|_{L^\infty(B_r(z))} \le Cr^{1+s+\tau}, \qquad Q_z(x) = a_z + b_z\cdot(x-z), \tag{2.9}$$

with bounded coefficients. Translate z to zero. If the estimate fails, take a sequence q_j, L_j with the fixed structural bounds. Let $Q_{j,r}(x) = a_{j,r} + b_{j,r}\cdot x$ be the least-squares affine fit to q_j/t^s on $B_r\cap\mathbb{H}$, that is, the minimizer of $\int_{B_r\cap\mathbb{H}} |q_j - Qt^s|^2$. The Gram matrix of $\{t^s, x_1 t^s, \dots, x_n t^s\}$ on the unit half-ball is positive definite. Define

$$\Theta(r) = \sup_j \sup_{\rho\ge r} \rho^{-1-s-\tau}\|q_j - Q_{j,\rho}t_+^s\|_{L^\infty(B_\rho)}.$$

The failure gives $\Theta(r)\to\infty$ as $r\downarrow 0$. Choose scales $r_j\downarrow 0$ and corresponding solutions for which the normalized error is at least $\Theta(r_j)/2$, and set

$$V_j(x) = \frac{q_j(r_j x) - Q_{j,r_j}(r_j x) r_j^s t_+^s}{r_j^{1+s+\tau}\Theta(r_j)}.$$

Finite-dimensional norm equivalence at two consecutive scales gives

$$|a_{j,r} - a_{j,2r}| \le C\Theta(r) r^{1+\tau}, \qquad |b_{j,r} - b_{j,2r}| \le C\Theta(r) r^{\tau}.$$

Telescoping from r_j to Rr_j therefore yields

$$\|V_j\|_{L^\infty(B_R)} \le C R^{1+s+\tau} \quad (R \ge 1), \qquad \|V_j\|_{L^\infty(B_1)} \ge \tfrac{1}{2}. \tag{2.10}$$

When Rr_j exceeds the original ball, boundedness of q_j and the same coefficient estimates give this bound as well.

The subtracted affine multiple of t_+^s has zero operator oscillation in $\mathbb{H}$ by (2.6). Thus the right-hand-side oscillation in the equation for V_j, on each fixed half-ball, is bounded by

$$C_R \frac{r_j^{\alpha-\tau}}{\Theta(r_j)} \longrightarrow 0.$$

The cutoff test (2.4) and the flat C^s estimate give compactness up to the flat boundary. After extracting the angular densities, (2.3) passes the equation to a locally uniform limit V. It vanishes on the lower half-space and has the growth (2.10). Lemma 2.1 makes it an affine polynomial times t_+^s. But least-squares orthogonality gives

$$\int_{B_1 \cap \mathbb{H}} V_j(x) Q(x) t^s \, dx = 0 \quad \text{for every affine } Q.$$

Passing to the limit and choosing its coefficient polynomial shows $V = 0$, contradicting (2.10). This proves (2.9); dyadic telescoping gives the limiting coefficients a_z, b_z and their boundedness.

It remains to recover the quotient's normal derivative. Let x_0 have height $d > 0$, let $z = (x_0', 0)$, and put $w = q - Q_z t_+^s$. Estimate (2.9), extended away from z by boundedness and the coefficient bounds, gives $|w(x)| \le C|x - z|^{1+s+\tau}$. Rescale at d and divide by $d^{1+s+\tau}$. On a fixed interior ball around e_n the remainder and its compensated tail are bounded, and the C^γ oscillation of its right-hand side is at most $Cd^{\alpha-\tau}$. After a fixed cutoff, (2.4) controls the remaining scalar. The interior estimate gives a uniform $C^{1+s+\alpha}$ norm there. Dividing by the smooth positive function t^s and scaling back gives, with $h = q/t^s$,

$$|h(x_0) - Q_z(x_0)| \le C d^{1+\tau}, \qquad |\nabla h(x_0) - b_z| \le C d^{\tau},$$

and the corresponding interior C^τ bound for ∇h. Comparing the affine approximations on overlapping half-balls gives $|b_z - b_{\widetilde{z}}| \le C|z - \widetilde{z}|^\tau$ for boundary points $z, \widetilde{z}$, and the compatible first-order estimate for $a_z, a_{\widetilde{z}}$. The interior and boundary cases together prove (2.8). □

3 Improvement of flatness for the actual solution

Put $p = 1 + s$ and $P_{\kappa,e}(x) = \kappa(x \cdot e)_+^p$. For a growing difference from such a profile, the tail that controls the oscillation of the operator is

$$\mathcal{T}(w) = \int_{|y|>1} |w(y)| \, |y|^{-n-2s-1} \, dy. \tag{3.1}$$

The extra power in this weight is dictated by (2.2). In particular $\mathcal{T}(P_{\kappa,e}) < \infty$, although its raw operator diverges.

We first recall the precise compactness input. For a bounded nonnegative solution of $\min\{Lw - f, w\} = 0$ in B_1, for an exponent $\vartheta \in (\max\{1, 2s\}, 1 + s)$ and $\sigma \in (0, s)$, [4, Lemma 6.5] gives a local C^q bound for some $q > \max\{1, 2s\}$ in terms of

$$[f]_{C^{\vartheta-2s}(B_1)} + \|w\|_{L^\infty_{1+s+\sigma}}. \tag{3.2}$$

The bound uses the forcing seminorm, without its constant value. We apply this estimate to finite cutoffs and rescalings of actual solutions. The classification [4, Theorem 1.5] says the following: if a globally nonnegative, locally Lipschitz and locally $C^{2s+\varepsilon}$ function V satisfies, for every $R \geq 1$ and $h \in \mathbb{R}^n \setminus \{0\}$,

$$\|\nabla V\|_{L^\infty(B_R)} \leq CR^{s+\sigma}, \quad L\left(\frac{V(\cdot + h) - V}{|h|}\right) \geq 0 \text{ in } \{V > 0\}, \quad 0 < \sigma < \min\{s, 1 - s\}, \tag{3.3}$$

then it is convex. For our homogeneous kernels, if $z_0 \in \partial\{V > 0\}$ and $L(\nabla V) = 0$ in its positive phase, then $V(x) = \kappa((x - z_0) \cdot e)^{1+s}_+$.

Proposition 3.1. *Fix $0 < \tau' < \tau < \alpha < \min\{s, 1 - s\}$ and $0 < \kappa_0 < \kappa_1 < \infty$. There are $\rho \in (0, 1/100)$ and $\delta_* > 0$, depending only on these parameters and the structural bounds, with the following property. Suppose v is bounded individually on $\mathbb{R}^n$, nonnegative there, locally C^q for some $q > \max\{1, 2s\}$, and*

$$\begin{gathered}\min\{Lv - f, v\} = 0 \text{ in } B_4, \qquad 0 \in \partial\{v > 0\}, \qquad [f]_{C^\gamma(B_4)} \leq \eta \leq \delta \leq \delta_*, \quad \delta > 0, \\ \kappa(t - \delta)^p_+ \leq v(x) \leq \kappa(t + \delta)^p_+ \quad (x \in B_2), \qquad \mathcal{T}(v - P_{\kappa,e_n}) \leq \delta, \qquad \kappa_0 \leq \kappa \leq \kappa_1,\end{gathered} \tag{3.4}$$

where $\gamma = 1 - s + \alpha$. Then there are κ', e' with $|\kappa' - \kappa| + |e' - e_n| \leq C\delta$ such that $v_\rho(x) = \rho^{-p} v(\rho x)$ satisfies

$$\begin{gathered}\kappa'(x \cdot e' - \delta')^p_+ \leq v_\rho(x) \leq \kappa'(x \cdot e' + \delta')^p_+ \quad (x \in B_2), \\ \mathcal{T}(v_\rho - P_{\kappa',e'}) \leq \delta', \qquad \delta' = \delta\rho^{\tau'}.\end{gathered} \tag{3.5}$$

The rescaled forcing seminorm is at most $\eta\rho^\alpha \leq \delta'$. The constants are independent of the individual global bound of v.

3.1 A strip decomposition and the forcing constant

We prove Proposition 3.1 by contradiction as $\delta \downarrow 0$. First we obtain estimates valid for each admissible v. Choose χ supported in B_2, equal to one on B_1, and put $P = \kappa t^p_+$ and $W = \chi(v - P)$. In the known positive half-ball $D_\delta = B_{1/2} \cap \{t > \delta\}$, direct subtraction of the operator at zero gives

$$LW = D + g, \qquad g(x) = f(x) - f(0) + \int (1 - \chi(y))(v - P)(y)(K(x - y) - K(-y))\, dy. \tag{3.6}$$

Here D is a scalar. At the actual free-boundary point zero, $Lv(0) = f(0)$ by continuity of the operator from positive points, since $q > 2s$. The compensated difference $LP(x) - LP(0)$ is zero for $t > 0$, by (2.6) and the derivative of P. Formula (2.2) and the tail assumption imply

$$\|g\|_{C^\gamma(D_\delta)} \leq C\delta. \tag{3.7}$$

The value bounds also imply, on the support of χ,

$$|W| \le C\delta^p \quad (|t| \le \delta), \qquad |W| \le C\delta(t+\delta)^s \quad (t \ge \delta), \qquad W = 0 \quad (t \le -\delta). \tag{3.8}$$

We need a bound for D, whose size is not controlled by the forcing seminorm alone. The harmonic influence of a thin exterior strip provides it. Exterior data bounded by $M_0\delta^s$ on $-\delta < t < \delta$ have harmonic extension to D_δ bounded by

$$E_\delta(t) = \frac{\sin(\pi s)}{\pi} M_0\delta^s (t-\delta)^s \int_{-\delta}^{\delta} \frac{(\delta - y)^{-s}}{t-y}\, dy, \qquad t > \delta. \tag{3.9}$$

On $t \le \delta$ this function is the strip datum $M_0\delta^s \mathbf{1}_{(-\delta,\delta)}$. The formula is the half-line Poisson formula. It follows from the one-dimensional interval formula [3, Proposition 1.10.10] by sending the right endpoint to infinity: the ratio

$$\frac{(t-\delta)(2R-t+\delta)}{(\delta-y)(2R+\delta-y)} \longrightarrow \frac{t-\delta}{\delta-y}$$

has an integrable majorant on the strip. The limit equation follows by bounded stability or direct verification. Formula (2.5) makes this a barrier for the same anisotropic operator. It satisfies

$$E_\delta \le M_0\delta^s, \qquad E_\delta(t) \le CM_0\delta\, t^{s-1} \quad (t \ge 2\delta). \tag{3.10}$$

Let τ_δ be the torsion function of D_δ, with zero exterior datum and $L\tau_\delta = 1$. At $x_\delta = 2\delta e_n$,

$$c\delta^s \le \tau_\delta(x_\delta) \le C\delta^s. \tag{3.11}$$

For the lower bound compare with the unit-forcing torsion function of a fixed-radius ball inside D_δ and tangent to its flat boundary. A fixed smooth interior bump gives a uniform positive interior lower bound. The Hopf estimate and its proof in [3, Proposition 2.6.6] then give the lower bound in (3.11). For the upper bound, a nonnegative smooth cutoff equal to one on B_1 is a supersolution after multiplication by a structural constant, so τ_δ is uniformly bounded. The flat boundary C^s estimate gives the desired upper bound.

All the auxiliary Dirichlet problems here exist in the weak energy class by [3, Theorem 2.2.24]; $n \ge 2$ and $s < 1$ imply $n > 2s$. The datum W is compactly supported and C^1 for each solution. Split its exterior values by smooth cutoffs into a strip part near zero and a part separated from the local flat boundary. The partitions have finite-energy extensions; in particular no artificial jump is made at the curved boundary. Weak comparison follows by testing the negative part in the symmetric energy form. The weak boundary estimate [3, Proposition 2.6.4] gives the continuous representatives before the continuous-solution boundary theorem or Proposition 2.2 is applied.

At x_δ, the harmonic extension of the strip part of W is $O(\delta^p)$, by (3.10) with $M_0 = C\delta$. The separated harmonic part has norm $O(\delta)$, vanishes on the local flat exterior, and is therefore also $O(\delta^p)$ there. The zero-exterior solution of $Lb = g$ is bounded in absolute value by $\|g\|_\infty \tau_\delta$. Representing W as these three terms plus $D\tau_\delta$, and using (3.8), gives

$$|D|\tau_\delta(x_\delta) \le C\delta^p + C\delta\tau_\delta(x_\delta).$$

It follows that

$$|D| \le C\delta. \tag{3.12}$$

3.2 Outer quotient and inner profile

Write $z = W/\delta$. Let e be the harmonic extension carrying its strip datum, with a smooth horizontal cutoff equal to one near zero and supported away from the curved boundary, and set $q_0 = z - e$. It is uniformly bounded, has zero local exterior on $t \le \delta$, and satisfies $Lq_0 = D/\delta + g/\delta$. Proposition 2.2, after translation and a fixed rescaling, gives

$$q_0(x', t) = (t - \delta)^s H_\delta(x', t - \delta), \qquad \|H_\delta\|_{C^{1,\tau}} \le C$$

on a fixed smaller half-ball. The strip bound gives $|e(x)| \le C\delta t^{s-1}$ for $t \ge 2\delta$. Hence

$$v(x) = \kappa t^p + \delta(t - \delta)^s H_\delta(x', t - \delta) + O(\delta^2 t^{s-1}), \qquad t \ge 2\delta. \tag{3.13}$$

Now let $\delta_j \downarrow 0$ and choose arbitrary actual free-boundary points $x_j = (x_j', \delta_j h_j)$ in a fixed small ball, so $|h_j| \le 1$. Pass to a subsequence with $x_j' \to x_*'$, $h_j \to h$ and $\kappa_j \to \kappa$. Center at x_j, rescale by δ_j, and divide values by δ_j^p. For applying (3.2), first multiply the actual v_j by a nonnegative fixed cutoff supported in B_2 and equal to one near all these points. Its rescaling agrees with the actual solution on every fixed ball eventually. The sandwich on the cutoff support gives a uniform global value bound

$$|V_j(y)| \le C(1 + |y|)^p. \tag{3.14}$$

The separated-tail forcing has Lipschitz oscillation bounded on a fixed rescaled ball by $C\delta_j^s$. This uses $\mathcal{T}(v_j - P_j) \le \delta_j$ and the finite first-order tail of P_j. The original forcing contributes at most $\eta_j \delta_j^\alpha$ to its C^γ seminorm. Thus (3.2) gives compactness in C^1_{loc} and in a lower C^q_{loc} with $q > 2s$. Let V be the nonnegative limit.

For every fixed R, dilate the finite V_j by R, divide by R^p, and apply (3.2) before letting j tend to infinity. The weighted value norm is bounded by (3.14), and the forcing oscillation tends to zero for that fixed R. This gives

$$\|\nabla V\|_{L^\infty(B_R)} \le CR^s. \tag{3.15}$$

It does not require an ordinary operator on the unbounded profile V. To pass complementarity to increments, let f_j be the forcing of the uncutoff v_j. Whenever $v_j(x) > 0$ and both $x, x + \delta_j h$ lie in B_4, the finite equation gives

$$L_j\big(v_j(\cdot + \delta_j h) - v_j\big)(x) \ge f_j(x + \delta_j h) - f_j(x).$$

There is equality if $v_j(x + \delta_j h) > 0$ as well. On each fixed inner ball the cutoff agrees with the actual rescaling. Its additional forcing has the vanishing separated-tail oscillation estimated above. After rescaling and division by $|h|$, the right-hand-side differences tend to zero for fixed $h \ne 0$. For the far part, transfer a signed increment to a difference of kernels with a smooth cutoff; its error is $O(|h|R^{-s})$ by the value growth. Passing first to the local limit and then sending the tail radius R to infinity gives the increment inequality in (3.3). On a compact set where both shifted points are positive, the limiting inequality is an equality. Sending the shift to zero gives $L(\nabla V) = 0$ in the positive phase, with the viscosity interpretation supplied by local interior regularity. The classification following (3.3) applies. The inherited slab bounds at every fixed radius force the resulting profile to have direction e_n and amplitude κ.

We must still retain the actual free-boundary point at the origin. A direct compensated calculation supplies this. For $t < 0$,

$$L_s(t_+^s) = -\mathrm{B}(s+1,s)(-t)^{-s}, \qquad \mathrm{B}(s+1,s) = \int_0^\infty z^s(1+z)^{-1-2s}\,dz > 0.$$

Differentiate large cutoffs of t_+^p, use their $O(R^{-s})$ derivative tail error, and integrate from zero to a fixed negative point. Since $p > 2s$, the resulting difference is continuous at zero. The compensated operator of $P_{\kappa,e}$ at signed distance $-d < 0$, relative to its positive-side value, is

$$A_e\kappa\frac{p\mathrm{B}(s+1,s)}{1-s}d^{1-s} > 0. \tag{3.16}$$

Here $A_e > 0$ is the one-dimensional factor defined in (2.5). If the origin were strictly inside the contact region of the limiting profile, choose positive points $y_j \to 0$ of V_j and a fixed positive point z of V. Local C^q convergence with $q > 2s$ and compensated tail convergence give

$$L_jV_j(y_j) - L_jV_j(z) \longrightarrow (LV)(0) - (LV)(z) \quad \text{in the compensated sense.}$$

The left side is a forcing difference and tends to zero. The right side is strictly positive by (3.16), a contradiction. Thus

$$V(y) = \kappa(y_n)_+^p. \tag{3.17}$$

The same calculation excludes actual positive pockets on any fixed compact set strictly below the limiting plane: a sequence of positive points in such a set would give the identical contradiction.

3.3 Signed matching and the improved value bounds

Extract a locally C^1 convergent subsequence of the uniformly $C^{1,\tau}$ functions H_{δ_j} in (3.13), with limit H_∞. Return to the arbitrary actual centers $x_j = (x_j', \delta_j h_j)$, with $x_j' \to x_*'$ and $h_j = x_{j,n}/\delta_j \to h$. Fix $T > 3$ and evaluate at the original point

$$x = x_j + \delta_j T e_n, \qquad t = \delta_j(T + h_j).$$

Divide the outer expansion (3.13) by the value scale δ_j^p. Its left side is $v_j(x_j + \delta_j T e_n)/\delta_j^p \to \kappa T^p$ by (3.17). Letting $j \to \infty$ at this fixed T gives

$$\kappa T^p = \kappa(T+h)^p + (T+h-1)^s H_\infty(x_*', 0) + O(T^{s-1}). \tag{3.18}$$

The error constant is uniform in $T > 3$. Only after taking $j \to \infty$ with T fixed do we subtract κT^p, divide by T^s, and send T to infinity. We obtain

$$H_\infty(x_*', 0) = -p\kappa h. \tag{3.19}$$

Because the free-boundary sequence was arbitrary, this matching is uniform on compact horizontal sets. At the original free-boundary point zero it gives $H_\infty(0,0) = 0$. Multiple components are included in this assertion.

Write $H_\infty(x) = b \cdot x + O(|x|^{1+\tau})$. Choose

$$\kappa_j' = \kappa_j + \delta_j b_n, \qquad e_j' = \frac{e_n + \delta_j b'/(p\kappa_j)}{|e_n + \delta_j b'/(p\kappa_j)|}. \tag{3.20}$$

The first variation of $P_{\kappa_j',e_j'}$ is $P_{\kappa_j,e_n} + \delta_j t^s(b\cdot x)$. The quotient Taylor remainder on $B_{2\rho}$ is bounded by $C\rho^{1+\tau}$. By (3.19), every actual free-boundary point lies within this normalized height error, plus $o(1)$, of the new plane. Choose ρ small enough that $C\rho^{\tau-\tau'}$ is small.

We verify value ordering in two regions. If $t \geq M_1\delta_j/\rho^{1+\tau'}$, the strip error in (3.13), divided by $\delta_j t^s$, is at most $C\delta_j/t \leq (C/M_1)\rho^{1+\tau'}$. With M_1 fixed sufficiently large, the quotient remainder and profile Taylor error are absorbed by the shifts $\delta_j\rho^{1+\tau'}$ in the original variables. The mean-value formula gives both value inequalities there.

In the remaining strip, choose actual free-boundary points on the vertical fibers. A top such point exists between heights $-\delta_j$ and δ_j. More explicitly, for a fixed small r_0 within the quotient chart, set $\Gamma_j = \partial\{v_j > 0\} \cap B_{r_0}$. For every fixed $R < \infty$, the unique centered limit (3.17) gives

$$\sup_{z\in\Gamma_j}\sup_{y\in B_R}\left|\frac{v_j(z+\delta_j y)}{\delta_j^p} - \kappa_j(y_n)_+^p\right| \longrightarrow 0.$$

A violating sequence of centers would have a subsequence with that same unique limit. Choose ρ first and then the finite inner radius $R = M_1/\rho^{1+\tau'}$, before sending j to infinity. The resulting value error is $o(\delta_j^p)$, smaller than the gap between the two positively shifted profiles, whose required nonzero gap is at least $c(\delta_j\rho^{1+\tau'})^p$.

For the zero region in the upper comparison one also needs actual contact, rather than small values. A positive pocket below the improved matching band would have a boundary point there: on its vertical fiber the solution is zero at height $-\delta_j$. This contradicts the arbitrary-point matching (3.19). The lower comparison is automatic where its model vanishes. These facts prove the genuine sandwich on $B_{2\rho}$ for large j, without an assumption of vertical monotonicity.

3.4 Tail improvement and iteration

Under the scaling in the proposition,

$$\mathcal{T}(v_\rho - P_{\kappa',e'}) = \rho^s\int_{|y|>\rho}|v(y) - P_{\kappa',e'}(y)||y|^{-n-2s-1}\,dy. \tag{3.21}$$

On each fixed annulus $\rho < |y| < r_0$, division by δ_j and passage to the limit gives $t_+^s(H_\infty(y) - b\cdot y)$. The original sandwich bounds the quotient through the vanishing strip, so dominated convergence applies. Its contribution to (3.21) is at most $C\delta_j\rho^\tau$, since $\tau < s$. Outside a fixed small r_0, the original sandwich, the tail assumption and the $O(\delta_j)$ profile change give at most $C\delta_j\rho^s$. Further decreasing ρ ensures

$$C(\rho^\tau + \rho^s) < \tfrac{1}{2}\rho^{\tau'}.$$

Thus the tail in (3.5) is at most $\delta_j\rho^{\tau'}$ for large j. Together with the value bounds this contradicts a sequence violating Proposition 3.1 and proves the existence of δ_*.

For iteration, fix $0 < \varepsilon < (\kappa_1 - \kappa_0)/2$ and start in the nonempty interval $[\kappa_0 + \varepsilon, \kappa_1 - \varepsilon]$, allowing the larger interval $[\kappa_0, \kappa_1]$. Decrease δ_* so that $C\delta_*/(1-\rho^{\tau'}) < \varepsilon$. Then $\delta_j = \delta_0\rho^{j\tau'}$, and the changes of amplitude and direction are summable. The value sandwiches imply

$$\|v\|_{L^\infty(B_r)} \leq Cr^p, \qquad 0 < r < 1. \tag{3.22}$$

To obtain the gradient, cut off the actual v in B_2. Its separated-tail forcing has uniformly Lipschitz oscillation by the original sandwich and $\mathcal{T}$, although its constant value need not be small. At scale r divide values by r^p. Estimate (3.22) bounds the global weighted value norm

of this cutoff. Its forcing seminorm is bounded by $C(\eta r^{\alpha} + r^{s})$. Applying (3.2) to each finite rescaling gives

$$\|\nabla v\|_{L^{\infty}(B_r)} \le C r^{s}, \qquad 0 < r < 1. \tag{3.23}$$

4 The interior estimate

We now return to the original obstacle problem and prove Theorem 1.1. Normalize

$$\|\phi\|_{C^{\beta}(B_1)} + \|u\|_{L^{\infty}(\mathbb{R}^n)} = 1. \tag{4.1}$$

If the left side is zero, the result is immediate. Choose, once and for all,

$$\begin{gathered} 0 < \alpha < \min\{\beta - 1 - s, s, 1 - s\}, \qquad b = 1 + s + \alpha < \min\{\beta, 1 + 2s, 2\}, \\ \gamma = 1 - s + \alpha, \qquad 0 < \sigma < \alpha, \qquad 0 < \tau' < \tau < \alpha. \end{gathered} \tag{4.2}$$

All these choices depend only on s, β.

4.1 Localization of the rough obstacle

The subcritical part of [4, Corollary 1.3] gives a uniform C^{q} bound in $B_{15/16}$ for a fixed $\max\{1, 2s\} < q < 1 + s$. Only that subcritical statement is used here. Take $\xi \in C_c^{\infty}(B_1)$ equal to one on $B_{15/16}$ and set $\widetilde{\phi} = \xi\phi$, extended by zero. Its global C^{b} norm is uniformly bounded. With $0 \le \chi \le 1$ supported in $B_{7/8}$ and equal to one on $B_{3/4}$, put

$$v = \chi(u - \widetilde{\phi}). \tag{4.3}$$

It is globally nonnegative, bounded and Lipschitz, with uniform bounds. In $B_{2/3}$ it agrees with $u - \phi$ and solves

$$\min\{Lv - f, v\} = 0, \qquad f = -L\widetilde{\phi} - L((1 - \chi)(u - \widetilde{\phi})), \qquad \|f\|_{C^{\gamma}(B_{2/3})} \le C. \tag{4.4}$$

The mapping estimate [3, Lemma 2.2.6] gives the first term's C^{γ} bound, since $b = 2s + \gamma$. The second term is a separated tail; differentiate its kernel once to obtain a Lipschitz bound, with no derivative of the exterior data. The local exponent $q > 2s$ makes the equation hold both classically and distributionally. At every zero of v, $\nabla v = 0$.

4.2 Coarse entry at a nondegenerate scale

We record the compactness consequence needed for the scale selection. Given $c_0 > 0$ and $\delta_0 > 0$, there is $\eta_0 > 0$ such that a bounded, globally nonnegative solution with

$$\begin{gathered} \min\{Lv - f, v\} = 0 \text{ in } B_{1/\eta}, \quad 0 \in \partial\{v > 0\}, \quad [f]_{C^{\gamma}} \le \eta \le \eta_0, \\ \|\nabla v\|_{L^{\infty}(B_R)} \le R^{s+\sigma} \quad (R \ge 1), \qquad \|\nabla v\|_{L^{\infty}(B_1)} \ge c_0 \end{gathered} \tag{4.5}$$

has a profile $P_{\kappa,e}$ with amplitude bounded above and bounded away from zero, and satisfies

$$\kappa(x \cdot e - \delta_0)_+^{p} \le v(x) \le \kappa(x \cdot e + \delta_0)_+^{p} \quad (x \in B_2), \qquad \mathcal{T}(v - P_{\kappa,e}) \le \delta_0. \tag{4.6}$$

The amplitude bounds depend only on c_0 and the structural parameters.

Indeed, otherwise let $\eta_j \downarrow 0$. The gradient growth in (4.5) gives weighted value growth of degree $p+\sigma$. The compactness estimate (3.2) gives a locally C^1 limit with a local exponent greater than $2s$. Complementarity passes to the signed increment inequalities as in the inner argument above. The classification (3.3) makes the limit a nonzero half-space profile. The compensated cusp argument (3.16) retains the origin as its actual free-boundary point and excludes positive points in negative compact sets. Thus it is $\kappa(x\cdot e)_+^p$, with a fixed positive lower amplitude from c_0.

In particular $v_j = 0$ on $B_2 \cap \{x\cdot e \le -\delta_0/4\}$ for large j. Uniform value convergence, with error $o(\delta_0^p)$, and the positive lower amplitude give the shifted comparisons (4.6). Where the lower profile vanishes its inequality is automatic. In the remaining region its positive shift gap, and likewise the upper shift gap above the actual zero region, dominate that error. For the tail, convergence on fixed annuli is local convergence, while the part beyond R is bounded by $CR^{\sigma-s}$. Since $\sigma < s$, first choose R large, then j large. This proves the coarse assertion.

Choose δ_0 below the threshold of Proposition 3.1 for comfortable nested amplitude intervals, and then $\eta_0 \le \delta_0$ accordingly. Rotating e to e_n preserves all structural bounds. The iteration leading to (3.23) gives

$$\|\nabla v\|_{L^\infty(B_r)} \le Cr^s \qquad (0<r<1) \tag{4.7}$$

whenever (4.5) holds.

4.3 First failure of subcritical decay

Fix an arbitrary actual free-boundary point $z \in B_{5/8}$ and translate it to zero. There is a fixed $R_{\mathrm{loc}} > 0$ on which (4.4) holds, uniformly in z. Let C_1 bound the global Lipschitz norm of v and C_f its local forcing seminorm. Choose $r_* \le R_{\mathrm{loc}}\eta_0/2$ and then A sufficiently large that

$$\frac{C_1}{Ar_*^{s+\sigma}} \le 1, \qquad \frac{C_f r_*^{\alpha-\sigma}}{A} \le \eta_0. \tag{4.8}$$

For $r_k = r_* 2^{-k}$ define

$$w_k(x) = \frac{v(r_k x)}{Ar_k^{1+s+\sigma}}, \qquad f_k(x) = \frac{r_k^{s-1-\sigma}}{A} f(r_k x). \tag{4.9}$$

Their equations hold in B_{1/η_0} and

$$[f_k]_{C^\gamma} \le C_f r_k^{\alpha-\sigma}/A \le \eta_0.$$

The strict inequality $\alpha > \sigma$ is used here. Let G_k denote

$$\|\nabla w_k\|_{L^\infty(B_R)} \le R^{s+\sigma} \quad \text{for every } R \ge 1. \tag{4.10}$$

By (4.8), G_0 holds. If every G_k holds, scaling back gives $\|\nabla v\|_{L^\infty(B_r)} \le Cr^{s+\sigma}$ and hence the endpoint estimate.

Otherwise let $K+1$ be the first failing index. The identity

$$\nabla w_{K+1}(x) = 2^{s+\sigma}\nabla w_K(x/2)$$

and G_K already give G_{K+1} for $R \ge 2$. If $\|\nabla w_K\|_{L^\infty(B_1)} \le 2^{-s-\sigma}$, they give it also for $1 \le R \le 2$. Thus

$$\|\nabla w_K\|_{L^\infty(B_1)} > c_0 = 2^{-s-\sigma}.$$

Apply the coarse assertion with this fixed c_0, followed by (4.7). For $r < r_K$ it yields

$$\|\nabla v\|_{L^\infty(B_r)} \le ACr_K^\sigma r^s \le ACr_*^\sigma r^s.$$

For $r_K \le r \le r_*$, G_K gives $\|\nabla v\|_{L^\infty(B_r)} \le Ar^{s+\sigma} \le Ar_*^\sigma r^s$. Larger fixed radii are controlled by the original Lipschitz bound. We have therefore proved, uniformly at every actual free-boundary point $z \in B_{5/8}$,

$$\|\nabla(u-\phi)\|_{L^\infty(B_r(z))} \le Cr^s \tag{4.11}$$

for all sufficiently small fixed radii. No nondegeneracy hypothesis remains.

4.4 From the one-point estimate to the full norm

Set $E = \{u = \phi\} \cap B_{2/3}$. Take a smooth cutoff θ supported in $B_{3/4}$ and equal to one on $B_{2/3}$, and let $U = \theta u$. The subcritical estimate bounds $W_i = \partial_i U$ globally. On the actual positive phase inside $B_{2/3}$,

$$LU = F_{\text{tail}} = -L((1-\theta)u), \qquad LW_i = \partial_i F_{\text{tail}} \tag{4.12}$$

in distributions. The separated tail is uniformly Lipschitz by one kernel derivative, so the last right-hand side is bounded. We have differentiated neither $L\phi$ nor the exterior data.

At a contact point z, $W_i(z) = \partial_i\phi(z)$. At a free-boundary point, (4.11) and $\nabla\phi \in C^{s+\alpha}$ give

$$|W_i(y) - \partial_i\phi(z)| \le C|y-z|^s \tag{4.13}$$

near z. For $x, y \in B_{1/2}$ put $h = |x-y|$. If x is in the positive phase and h is comparable to or greater than $d = \operatorname{dist}(x, E)$, a closest contact point and (4.13) give the Ch^s bound for small d, h. For distances bounded below the global derivative bound suffices.

Suppose next h is much smaller than d and d is small. Choose a closest contact point z, which is an actual interior free-boundary point, and put

$$Z_i(X) = \frac{W_i(z+dX) - \partial_i\phi(z)}{d^s}.$$

Estimate (4.13) near z and boundedness away from z imply the uniform weighted value growth $|Z_i(X)| \le C(1+|X|)^s$. On a fixed interior ball around $(x-z)/d$, its equation has bounded right-hand side $d^s\partial_i F_{\text{tail}}$. Choose a finite integrability exponent p_0 so that

$$s < \mu = 2s - n/p_0 < \min\{2s, 1\}.$$

The interior estimate [3, Theorem 2.4.3] gives a uniform C^μ bound there; the tail growth $s < 2s$ is admissible. Consequently

$$|W_i(x) - W_i(y)| \le Cd^s(h/d)^\mu \le Ch^s.$$

If d is bounded below, u is harmonic on a fixed ball and the interior estimate [3, Theorem 2.4.1], with the regular kernel tail, gives the same conclusion. If both points lie in E, use $\nabla u = \nabla\phi$. For a mixed pair with a point in the interior of E, the segment to the positive point meets an actual free-boundary point within distance h of both. Apply (4.11) there and then the obstacle-gradient modulus. If the coincidence set is locally empty, the harmonic case applies.

These cases cover $B_{1/2}$ and show $[\nabla u]_{C^s(B_{1/2})} \le C$. The supremum norms of u and ∇u were already bounded by the normalization and the subcritical estimate. Restoring (4.1) proves

$$\|u\|_{C^{1+s}(B_{1/2})} \le C\big(\|\phi\|_{C^\beta(B_1)} + \|u\|_{L^\infty(\mathbb{R}^n)}\big),$$

where C depends only on $n, s, \beta, \lambda, \Lambda$ and $\|a\|_{\text{Lip}(S^{n-1})}$, as asserted. □

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